

A FAST ALGORITHM TO DETERMINE FRACTAL DIMENSIONS BY BOX COUNTING

Larry S. LIEBOVITCH and Tibor TOTH

Department of Ophthalmology, Columbia University, 630 West 168th Street, New York, NY 10032, USA

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A new algorithm is used to determine fractal dimensions by box counting for dynamic and iterated function systems. This method is fast, accurate, and less dependent on data specific curve fitting criteria than the correlation dimension.

The fractal dimension of a set in a metric space such as a geometric object or the phase space trajectory of a dynamical system can be computed from several different measures including: the first three generalized dimensions [1], the capacity [2-4], the information dimension [5], and the correlation dimension [6]; the Liapunov exponents [7,8]; and the singular value decomposition [9]. These methods are reviewed in refs. [10-12].

Recently, one of the most used measures has been the spatial correlation dimension computed from the algorithm of Grassberger and Procaccia [6]. The correlation dimension d_C is defined by

$$d_C = d \log C(r) / d \log r \quad (1)$$

for small $r > 0$, where $C(r)$, the correlation integral, is given by

$$C(r) = \lim_{N \rightarrow \infty} \frac{1}{N^2} \sum_{j=1}^N \sum_{i=j+1}^N \theta(r - \|R_i - R_j\|). \quad (2)$$

Here N is the number of points, θ is the Heaviside function, and the norm is the Euclidean one. In practice, however, the slope of $\log C(r)$ versus $\log r$, and thus d_C , determined from experimental data is not often a constant, but varies as a function of r [9,13,14]. One of the reasons this happens is that the correlation sum from points near the edge of the set is underestimated. Since there is no general procedure to choose the best estimate of d_C , the criteria to determine the fit of $\log C(r)$ versus $\log r$ depend on the details of each data set analyzed. Moreover, these computations can be time consuming. To com-

pute all pairs of distances increases as N^2 . This time can be reduced to order $N \log N$ by organizing the points into boxes [15]. Sometimes, only a small number, typically a few per cent, of "reference" points are chosen as R_i , which decreases the computational time at the expense of increasing the uncertainty in d_C [16].

An alternative measure of the dimension is the capacity [2,4,10,12,17]

$$d_B = \lim_{\epsilon \rightarrow 0} \log N_B(\epsilon) / \log(1/\epsilon). \quad (3)$$

This is called "box-counting" because one counts the minimal number of boxes, $N_B(\epsilon)$, that cover the set, for boxes of size ϵ . Useful algorithms to implement this were described by Grassberger [3], Hunt and Sullivan [18], Giorgilli et al. [19], and Theiler [15]. Barnsley [4] provides a number of examples demonstrating the utility of box counting to determine the fractal dimension in the plane. However, this method has not been widely used because it required: (1) too large a number of datapoints and (2) too much computer memory and computation time. We will show how these limitations can be overcome. Moreover, our method does not require the curve fitting decisions dependent on the details of each data set required by the use of the correlation dimension.

Greenside et al. [20] found that taking the limit in eq. (3), that is, using very small box sizes, required an impractically large number of datapoints. However, this criterion is more restrictive than nec-

essary. The fractal dimension describes how many new pieces of a set are resolved as the resolution scale is decreased [21]. Since a fractal is self-similar, this means that the fractal dimension can be evaluated by comparing properties between any two scales, namely,

$$d_B \approx d \log N_B(\epsilon) / d \log(1/\epsilon). \quad (4)$$

In practice, the correlation dimension is also evaluated using this fact by fitting $\log C(r)$ versus $\log r$ over an intermediate range of r values and excluding the regimes near $r \rightarrow 0$ and near $r \rightarrow r_{\max}$ where the accuracy of the determination of $C(r)$ is affected by the finite size of the data set. We will show below that the fractal dimension can be determined accurately from eq. (4), and present an efficient algorithm that requires only approximately Nd_e memory locations where d_e is the embedding dimension and executes in time of order $N \log N$.

This box counting method to determine the fractal dimension is particularly useful when the dimension is low. For example it is very efficient for the analysis of two-dimensional images such as the sets illustrated by Barnsley [4]. However, since it requires approximately 10^d data points, where d is the dimension of the attractor, it is less useful when the dimension is high. Note that other methods of determining the dimension may also be subject to a similar limitation, for example, Wolf et al. [7] note that their method, using Liapunov exponents, also requires 10^d or more data points.

To compute d_B , we need to count the number of boxes in a minimal cover that contain at least one element of the set. This is then carried out for a sequence of decreasing box sizes. Our algorithm does this by using an efficient hashing to code all the points within one box with the same number and then to count the number of distinct values.

Each of the N points of a set embedded in d_e dimensions can be represented by a vector with coordinates $\{X_i; i=1, d_e\}$. The values of X_i are normalized to cover the range $(0, 2^k-1)$. The set is covered by a grid of d_e dimensional cubes of edge size 2^m , $0 \leq m \leq k$, called boxes. For each coordinate we form $Y_i = (X_i \text{ AND } M)$ where AND is the binary conjunction of the corresponding bits in X_i and M , and M is a binary number with 1's in the first $k-m$

places and 0's the remainder. Then for each $n=1, \dots, N$ we construct $Z_n = Y_1 + Y_2 + Y_3 \dots + Y_{d_e}$, where the operation "+" indicates concatenation (for example, "10" + "01" = "1001"). All the points within the same box of size 2^m will have coordinates that have identical binary digits in the first $k-m$ places. Thus, distinct Z_n correspond to points in distinct boxes. We count the number of distinct Z_n efficiently by ordering them using a quicksort or heapsort [22] to order all N strings Z_n , and then walk down the list once to count the number of times the values change. The procedure is then repeated for different boxes of edge size 2^m , where $m=k, k-1, \dots, 0$.

We performed the above procedure in a BASIC program by means of the concatenation of alphanumeric variables (strings) to form Z_n , and a quicksort algorithm [23]. Note that the memory required is determined by the number of data points and not the number of boxes in the grid. The data points were two-byte integers, so that the total memory required was approximately $2Nd_e$ bytes, where N is the number of data points and d_e the embedding dimension. The computational time for this box counting algorithm depends mostly on the time required by the sort which is approximately $N \log N$. The computational time increases slowly with d_e , the details of which depend on the hardware used.

The limited resolution due to the finite number N of the data points is reached when ϵ is small enough so that all the points lie in distinct boxes, that is, when $N_B(\epsilon) = N$. At this value of ϵ the function $N_B(\epsilon)$ is saturated. Values of $N_B(\epsilon)$ near saturation should be disregarded when the slope of $\log N_B(\epsilon)$ versus $\log(1/\epsilon)$ is calculated. To be enough below this point, only values of $N_B(\epsilon) \leq N/5$ were taken into account. Moreover, since the largest box sizes have such poor resolution, we also ignore $N_B(\epsilon)$ for $m=k, k-1$. We then use a least squares fit to determine d_B as the slope of $\log N_B(\epsilon)$ versus $\log(1/\epsilon)$. Note, we always use the same procedure of fitting $N_B(\epsilon)$ versus $\log(1/\epsilon)$ from $m=k-2$ to $N_B(\epsilon) \leq N/5$. Thus, we always have a fixed procedure to determine this fit. This is a significant improvement over the evaluation of the correlation dimension, where the fit of $\log C(r)$ versus $\log r$ depends on the details of the particular data set analyzed that determine the range of r over which d_C is evaluated. However, if the data consisted of a union

of subsets of different dimensions, our procedure would lump them together.

To compare their accuracy and computational times, we computed the correlation dimension d_c by the Grassberger-Procaccia algorithm and the capacity dimension d_B by our new algorithm for two types of attractors. The first type was an example of a deterministic dynamic system. We used the Hénon map [24]:

$$x_{i+1} = 1 + y_i - 1.4x_i^2, \quad y_{i+1} = 0.3x_i. \quad (5)$$

The fractal dimension for this map is known from estimates based on Liapunov numbers [25], capacity dimension [3,19,25] and correlation dimension [6]. However, the fractal dimension cannot be determined analytically for this, or many other, deterministic dynamic systems. Thus, to test the accuracy of these methods we also applied them to analyze a second type of set where the fractal dimension can be evaluated analytically. Such fractals were constructed using the Collage theorem [4]. They were generated from a set of affine transformations applied in random order. The transformations are such that the union of their images approximates the fractal desired. For example, the middle third Cantor set which has fractal dimension $\log 2 / \log 3 \approx 0.631$ was formed by consecutive applications of the following set of affine transformations,

$$x_{i+1} = x_i/3, \quad x_{i+1} = 2/3 + x_i/3, \quad (6)$$

applied with equal probability to the result of the previous transformation. To produce a modified Sierpinski triangle map with fractal dimension $\log 3 / \log 2 \approx 1.58$ the set of three transformations ($k=1, 2, 3$)

$$x_{i+1} = y_i/2 + a_k, \quad y_{i+1} = x_i/2 + b_k, \quad (7)$$

where $a_1 = a_2 = b_1 = 1$ and $a_3 = b_2 = b_3 = 125$, were applied with equal probability. The Hénon and modified Sierpinski attractors are shown in fig. 1.

The Hénon map and the iterated function systems were used to compute N values of $\{x_i\}$ for $N=1000$ and 10000. In analyzing experimental data, typically the time series of only one of the variables is known. The attractor can then be reconstructed, in an embedding space of dimension d_e , from points whose coordinates are values of the time series separated by a constant lag time [26,27]. For the test cases here,

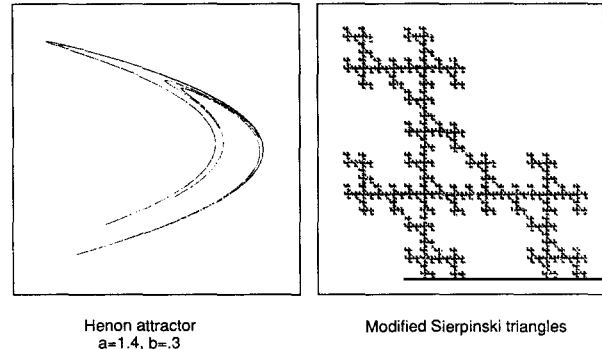


Fig. 1. Plot of $\{x_i, y_i\}$ for the Hénon attractor of eq. (5) and the iterated function system representing the modified Sierpinski triangle map described by eq. (7).

the correlation length is one iteration, and thus the points $(x_i, x_{i+1}, x_{i+2}, \dots, x_{i+d_e-1})$ for $i=1$ to $N-d_e+1$ were used to reconstruct the trajectories. Embedding dimensions from $d_e=3$ to $d_e=11$ were used. The correlation function $C(r)$ versus r and the number of boxes of size ϵ occupied by at least one point of the trajectory $N_B(\epsilon)$ were then computed. The dimensions were determined as the least squares estimates of

$$d_c = \frac{d \log C(r)}{d \log r} \quad \text{and} \quad d_B = \frac{d \log N_B(\epsilon)}{d \log (1/\epsilon)}.$$

These results are shown in fig. 2 for the Hénon map and in fig. 3 for the modified Sierpinski triangle map. Note that d_B is a function of $1/\epsilon$, but that its average over different scales, always using the same fixed range from $m=k-2$ to $N_B(\epsilon)=N/5$, yields an accurate estimate of the fractal dimension. The estimates of the dimensions are given in table 1 when the embedding dimension $d_e=3$ and the number of points $N=1000$ and 10000. All computations were done using an Apple Macintosh II and compiled Microsoft® QuickBASIC.

We have described a simple, fast, and accurate box counting method to determine the fractal dimension. This method has a fixed rule to fit the experimental data. This is an advantage over the correlation dimension where the range of r chosen to evaluate the dimension will vary with each particular data set. This box counting method is particularly well suited for low dimensional systems, such

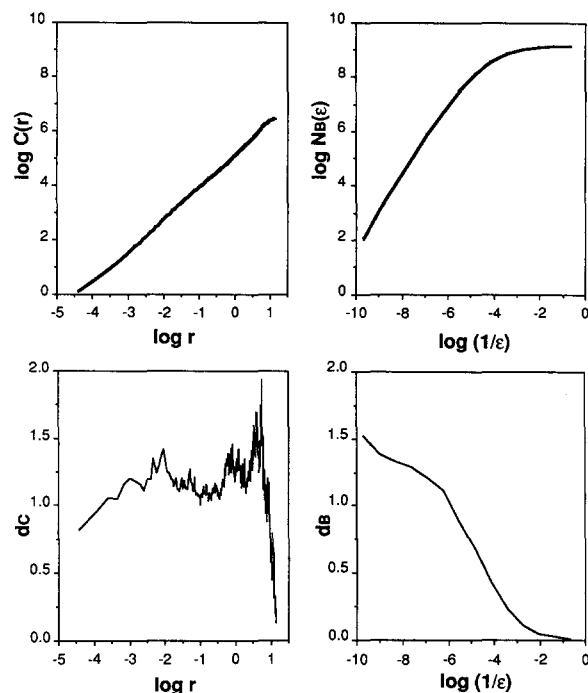


Fig. 2. Shown for the Hénon map with $d_c=3$ and $N=10000$ are the correlation function $C(r)$, and the number of boxes $N_B(\epsilon)$, of size ϵ occupied by the set. Below, are the correlation dimension $d_C = d \log C(r) / d \log r$ and the capacity dimension $d_B = d \log N_B(\epsilon) / d \log(1/\epsilon)$.

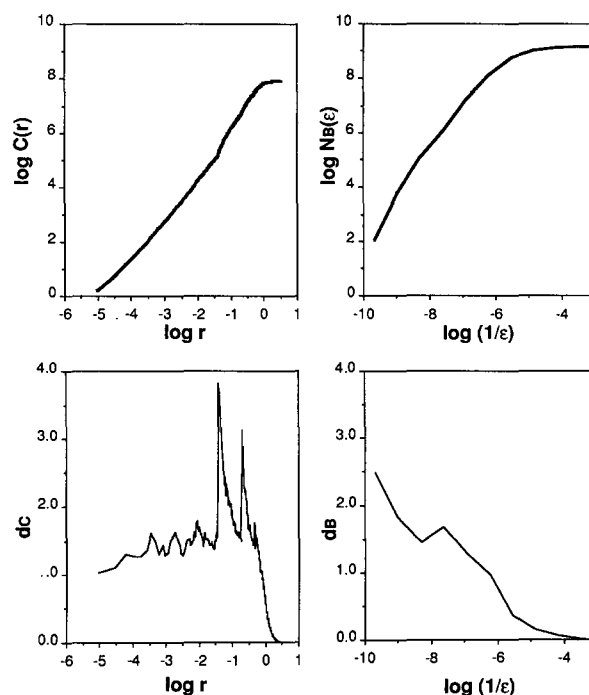


Fig. 3. Shown for the iterated function system representing the modified Sierpinski triangle map with $d_c=3$ and $N=10000$ are the correlation function $C(r)$, and the number of boxes $N_B(\epsilon)$, of size ϵ occupied by the set. Below, are the correlation dimension $d_C = d \log C(r) / d \log r$ and the capacity dimension $d_B = d \log N_B(\epsilon) / d \log(1/\epsilon)$.

as the analysis of two-dimensional images.

Note added. Daniel Kaplan has suggested to us a revision of our algorithm that considerably reduces

the execution time by using binary operations. If the high to low ordering of the bits in Z_n is the cyclic ordering of the bits in the coordinates Y_i , then the Z_n

Table 1

The results displayed correspond to embedding dimension 3. The Grassberger–Procaccia algorithm was used with reference points being 1% of the total amount of data points. The errors stated are those of the least-squares fit, which underestimate the systematic errors in both methods.

	Exact fractal dimension	Correlation dimension by Grassberger–Procaccia algorithm (computational time in minutes)		Capacity dimension by new box counting algorithm (computational time in minutes)	
		$N=1000$	$N=10000$	$N=1000$	$N=10000$
Hénon attractor	1.26	1.15 ± 0.01 (2)	1.21 ± 0.002 (420)	1.32 ± 0.02 (2)	1.27 ± 0.02 (19)
modified Sierpinski	1.58	1.50 ± 0.03 (2)	1.53 ± 0.01 (580)	1.62 ± 0.001 (2)	1.63 ± 0.04 (17)
middle third Cantor set	0.631	0.611 ± 0.012 (2)	0.630 ± 0.003 (570)	0.701 ± 0.015 (2)	0.639 ± 0.007 (16)

need to be sorted only once for all box sizes. Then for each box size, the box size dependent masking is performed, and the list of Z_n searched for distinct numbers.

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